RETAIL PERFORMANCE ANALYSIS

A Business Intelligence project analysing retail sales performance across products, customers and regions using Excel, SQL and Microsoft Power BI.

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1. Business Background

The retail company sells a wide range of products across multiple categories and regions. While the company has maintained consistent sales performance, management seeks a deeper understanding of business performance to identify opportunities to improve profitability, optimise operations and support strategic decision-making.

2. Business Problem

Although the company continues to generate consistent sales, management has limited visibility into the factors influencing profitability. As a result, it is difficult to identify which products, customer segments and regions contribute most to financial performance, where underperformance exists and which areas present the greatest opportunities for improvement.

3. Project Objective

To analyse retail sales data to evaluate profitability across products, customers and regions, identify key performance drivers, and develop data-driven insights and recommendations that support informed business decision-making and improve overall organisational performance.

4. Business Questions

From the business problem many questions are needed to be investigated and answered to help achieve the objective.

These questions are:
1. How has profit changed over time?
2. Which product categories, sub-categories, and products generate the highest profits?
3. Which regions generate the highest profits, and which regions underperform?
4. Which customer segments and individual customers generate the highest profits?
5. Which regional manager oversees the highest-profit region?
6. Which regions have the highest return rates, and how do returns affect profitability across regional managers?

5. Dataset Overview

The dataset used for this project was sourced from Kaggle and is provided as an Excel (.xlsx) file titled "Superstore Sales Analysis". It consists of three tables: Orders (10,194 records, 21 columns), People (4 records, 2 columns), and Returns (296 records, 2 columns). The Orders' table contains transactional sales and customer information. The People's table contains information on the regional manager. The Returns' table contains information on whether the order has been returned.

The dataset contains transactional retail sales data for a fictional superstore business covering the period from 2023 to 2026. It includes detailed information on customer orders, product categories, shipping methods, sales, discounts, profits, and returns, enabling analysis of sales performance, profitability, customer behaviour, and regional trends.

6. Data Preparation

The dataset was assessed for data quality prior to analysis. Checks were performed for missing values, duplicate records, data type consistency, and formatting issues. No significant data quality issues were identified, so only minimal preparation was required before importing the data into SQL.

7. Methodology

The project followed a structured data analytics workflow. The dataset was initially explored and reviewed in Microsoft Excel to assess its quality and structure. After confirming the data was suitable for analysis, it was imported into MySQL where SQL queries were developed to answer the key business questions. The processed data was then connected to Microsoft Power BI, where relationships between the tables were established and DAX measures were created to calculate key performance indicators, including Total Sales, Total Profit, Profit Margin, Total Orders and Return Rate. Finally, three interactive dashboards were developed to present insights into executive performance, product performance, and customer and regional performance.

8. SQL Result

8.1. How has sales and profit changed over time?

Using the SQL query in Appendix A.1, both total sales and total profit generally increased over the period from 2023 to 2026. Total sales increased from $494,040.21 in 2023 to $745,567.53 in 2026, while total profit increased from $51,684.30 to $95,926.35 during the same period.

8.2. Which product categories, sub-categories, and products generate the highest profits?

Using the SQL query in Appendix A.2.1, A.2.2, A.2.3, the Technology category generated the highest total profit ($146,543.38). Among the sub-categories, Copiers recorded the highest total profit ($56,093.94), while the Canon imageCLASS 2200 Advanced Copier generated the highest total profit ($25,199.93) of all individual products.

8.3. Which regions generate the highest profits, and which regions underperform?

Using the SQL query in Appendix A.3, the West region generated the highest total profit ($110,798.82), while the Central region recorded the lowest total profit ($39,865.31).

8.4. Which customer segments and individual customers generate the highest profits?

Using the SQL queries in Appendices A.4.1 and A.4.2, the Consumer segment generated the highest total sales and total profit among all customer segments, with a total profit of $136,371.45. At the individual customer level, Tamara Chand generated the highest total profit of $8,981.32.

8.5. Which regional manager oversees the highest-profit region?

Using the SQL query in Appendix A.5, Sadie Pawthorne, the regional manager for the West region, was associated with the highest total sales and total profit, with total profit reaching $110,798.82.

8.6. Which regions have the highest return rates, and how do returns affect profitability across regional managers?

Using the SQL query in Appendix A.6, the West region recorded the highest return rate (11.56%). Despite this, the region also generated the highest total sales ($739,813.61) and total profit ($110,798.82).

9. Power BI Dashboard Analysis

An interactive version of the dashboards presented in this section is available by downloading the retail_analysis_dashboard.pbix file from the retail-performance-analysis GitHub repository and opening it in Microsoft Power BI Desktop.

GitHub Repository: https://github.com/oarkhonglim/retail-performance-analysis

9.1 Executive Summary Dashboard

The Executive Dashboard (Appendix B.1) provides management with a high-level overview of the organisation's financial performance between 2023 and 2026. Interactive slicers enable users to filter the dashboard by year, month and region, allowing business performance to be analysed across different time periods and geographic locations.

The dashboard supports strategic decision-making by presenting key performance indicators (KPIs), including Total Sales, Total Profit, Profit Margin and Total Orders. Trend analysis enables management to monitor business growth over time, while regional profit comparisons help identify the strongest and weakest performing markets.

The dashboard highlights several important business insights. Total sales increased from $494,040.21 in 2023 to $745,567.53 in 2026, while total profit grew from $51,684.30 to $95,926.35, indicating sustained business growth over the four-year period. The West region generated the highest total profit ($110,798.82), demonstrating that it is the organisation's strongest performing market. In contrast, the Central region generated the lowest total profit ($39,865.31), highlighting an opportunity for management to investigate the factors contributing to lower profitability. Understanding the strategies contributing to the West region's success may help improve performance in underperforming regions through the adoption of best practices and more effective resource allocation.

9.2 Product Performance Dashboard

The Product Performance dashboard (Appendix B.2) enables management to evaluate profitability across product categories, sub-categories and individual products. Interactive filters allow users to analyse product performance across different time periods while comparing categories and sub-categories to identify trends and areas requiring attention.

The dashboard supports product portfolio management by identifying high-performing and underperforming products. Comparing sales and profit across categories enables decision-makers to assess whether strong sales are translating into strong profitability, while the sub-category analysis highlights products that negatively impact overall business performance.

The dashboard identified Technology as the highest-performing category, generating total sales of $839,893.28 and total profit of $146,543.38. While all three product categories generated positive profits overall, the Supplies (-$1,171.39), Bookcases (-$3,632.07) and Tables (-$17,753.21) sub-categories recorded overall losses. These findings suggest that pricing strategies, discount policies, product costs and customer demand may be contributing factors to the negative profitability observed within these sub-categories.

At the individual product level, the Canon imageCLASS 2200 Advanced Copier generated the highest total profit ($25,199.93). This indicates that products generating the highest profits play a significant role in overall business performance and may present opportunities for increased marketing investment, inventory prioritisation or product expansion to maximise future profitability.

9.3 Customers & Regions Dashboard

The Customer & Regional Insights dashboard (Appendix B.3) provides management with an overview of customer profitability, regional performance and product returns. Interactive filters enable users to compare customer segments, identify high-value customers and evaluate regional performance across different time periods and locations.

The dashboard supports customer relationship management by identifying the customer segments and individual customers that contribute most to organisational profitability. In addition, regional manager performance and return rate analysis provide operational insights that can assist management in evaluating regional performance and identifying opportunities to reduce costs associated with product returns.

The analysis shows that the Consumer segment contributed the largest share of total profit, accounting for 46.66% ($136,371.45) of overall profitability, making it the organisation's most valuable customer segment. At the individual customer level, Tamara Chand generated the highest total profit ($8,981.32). Regionally, Sadie Pawthorne, who manages the West region, achieved the highest regional profit ($110,798.82).

Although the West region also recorded the highest return rate (11.56%), it remained the organisation's most profitable region. This suggests that strong sales performance continues to offset the financial impact of returned orders. However, the comparatively higher return rate indicates an opportunity for management to investigate potential causes, such as product quality, customer expectations or fulfilment processes, to further improve profitability and operational efficiency.

10. Recommendation

Recommendation 1: Expand investment in Technology products

The Technology category generated the highest sales ($839,893.28) and profit ($146,543.38), demonstrating consistently strong financial performance.

Management should consider increasing investment in high-performing technology products through targeted marketing campaigns, inventory planning and product portfolio expansion to further increase sales and profitability.

Recommendation 2: Review loss-making sub-categories

The Tables, Bookcases and Supplies sub-categories recorded overall losses despite generating sales, indicating that revenue is not translating into profitability.

Management should review pricing strategies, discount policies, supplier costs and product demand to identify the causes of negative profitability. Where improvements cannot be achieved, the organisation should consider discontinuing consistently unprofitable products.

Recommendation 3: Investigate product returns in the West region

The West region generated the highest total profit but also recorded the highest return rate (11.56%), suggesting opportunities to improve operational performance.

Management should investigate the underlying causes of product returns, including product quality, fulfilment processes and customer expectations. Reducing returns while maintaining strong sales performance could further improve regional profitability, customer satisfaction and the organisation's brand reputation.

Recommendation 4: Apply successful regional strategies across the organisation

The West region significantly outperformed the Central, East and South regions in terms of profitability.

Management should analyse the sales strategies, operational practices and customer engagement approaches used within the West region to determine whether these practices can be implemented across the remaining regions to improve overall organisational performance.

Recommendation 5: Strengthen customer retention within the Consumer segment

The Consumer segment contributed 46.66% of total profit, making it the organisation's most valuable customer segment.

Management should strengthen customer retention through targeted marketing campaigns, loyalty programmes and personalised promotions. Retaining high-value customers is likely to increase customer lifetime value and support long-term profitability.

11. Conclusion

This project successfully demonstrated how SQL and Microsoft Power BI can be used to transform retail sales data into meaningful business insights that support strategic decision-making. SQL was used to answer key business questions relating to profitability, product performance, customer behaviour and regional performance, while Power BI presented these findings through interactive dashboards that enable users to explore business performance using dynamic filters and visualisations.

The analysis identified Technology as the highest-performing product category, while the Canon imageCLASS 2200 Advanced Copier generated the highest individual product profit. The Consumer segment contributed the largest share of overall profitability, and the West region consistently outperformed the other regions despite recording the highest return rate. These findings highlight both the organisation's key strengths and areas for improvement, particularly in reducing product returns, improving the profitability of underperforming sub-categories and applying successful regional strategies across the organisation.

Overall, this project demonstrates the value of combining SQL for data analysis with Power BI for interactive business reporting. The insights and recommendations developed from the analysis provide management with practical, evidence-based guidance to support strategic decision-making, improve operational performance and enhance long-term profitability.

12. Appendix

Appendix A: SQL Queries

A.1 Profit Over Time

```
SELECT Year(`Order Date`) AS OrderYear, SUM(Sales) AS TotalSales,
SUM(Profit) AS TotalProfits
FROM Orders
GROUP BY OrderYear
ORDER BY OrderYear;
```

A.2.1 Category Contributing to Sales and Profit

```
SELECT Category, SUM(Sales) As TotalSales, SUM(Profit) AS TotalProfits
FROM Orders
GROUP BY Category
ORDER BY TotalProfits DESC, TotalSales DESC;
```

A.2.2 Sub-Category Contributing to Sales and Profit

```
SELECT `Sub-Category` AS SubCategory, SUM(Sales) As TotalSales,
SUM(Profit) AS TotalProfits
FROM Orders
GROUP BY SubCategory
ORDER BY TotalProfits DESC, TotalSales DESC;
```

A.2.3 Product Contributing to Sales and Profits

```
SELECT `Product Name` AS ProductName, SUM(Sales) As TotalSales,
SUM(Profit) AS TotalProfits
FROM Orders
GROUP BY ProductName
ORDER BY TotalProfits DESC, TotalSales DESC;
```

A.3 Regional Profitability

```
SELECT Region, SUM(Sales) AS TotalSales, SUM(Profit) AS TotalProfits
FROM Orders
GROUP BY Region
ORDER BY TotalProfits DESC;
```

A.4.1 Customer Segment Profitability

```
SELECT Segment, SUM(Sales) AS TotalSales, SUM(Profit) AS TotalProfits
FROM Orders
GROUP BY Segment
ORDER BY TotalProfits DESC, TotalSales DESC;
```

A.4.2 Top Customer Profitability

```
SELECT `Customer Name` AS CustomerName, SUM(Sales) AS TotalSales,
SUM(Profit) AS TotalProfits
FROM Orders
GROUP BY CustomerName
ORDER BY TotalProfits DESC, TotalSales DESC;
```

A.5 Regional Manager Profitability

```
SELECT p.`Regional Manager` AS RegionalManager, p.Region, SUM(o.Sales)
AS TotalSales, SUM(o.Profit) AS TotalProfits
FROM Orders o
JOIN people p
ON o.Region = p.Region
GROUP BY RegionalManager, p.Region
ORDER BY TotalProfits DESC, TotalSales DESC;
```

A.6 Return Rate Analysis

```
#DISTINT is used on Order ID as multiple products can link to the same
Order ID

SELECT
    p.`Regional Manager` AS RegionalManager,
    p.Region,
    COUNT(DISTINCT o.`Order ID`) AS TotalOrders,
    COUNT(DISTINCT CASE WHEN r.Returned = 'Yes' THEN o.`Order ID` END)
AS TotalReturned,
    ROUND(COUNT(DISTINCT CASE WHEN r.Returned = 'Yes' THEN o.`Order ID`
END) * 100.0 / COUNT(DISTINCT o.`Order ID`), 2) AS ReturnRate,
    SUM(o.Sales) AS TotalSales,
    SUM(o.Profit) AS TotalProfits
FROM Orders o
LEFT JOIN People p ON o.Region = p.Region
LEFT JOIN Returns r ON o.`Order ID` = r.`Order ID`
GROUP BY p.`Regional Manager`, p.Region
ORDER BY ReturnRate DESC;
```

Appendix B: Power BI Dashboards

B.1 Executive Dashboard

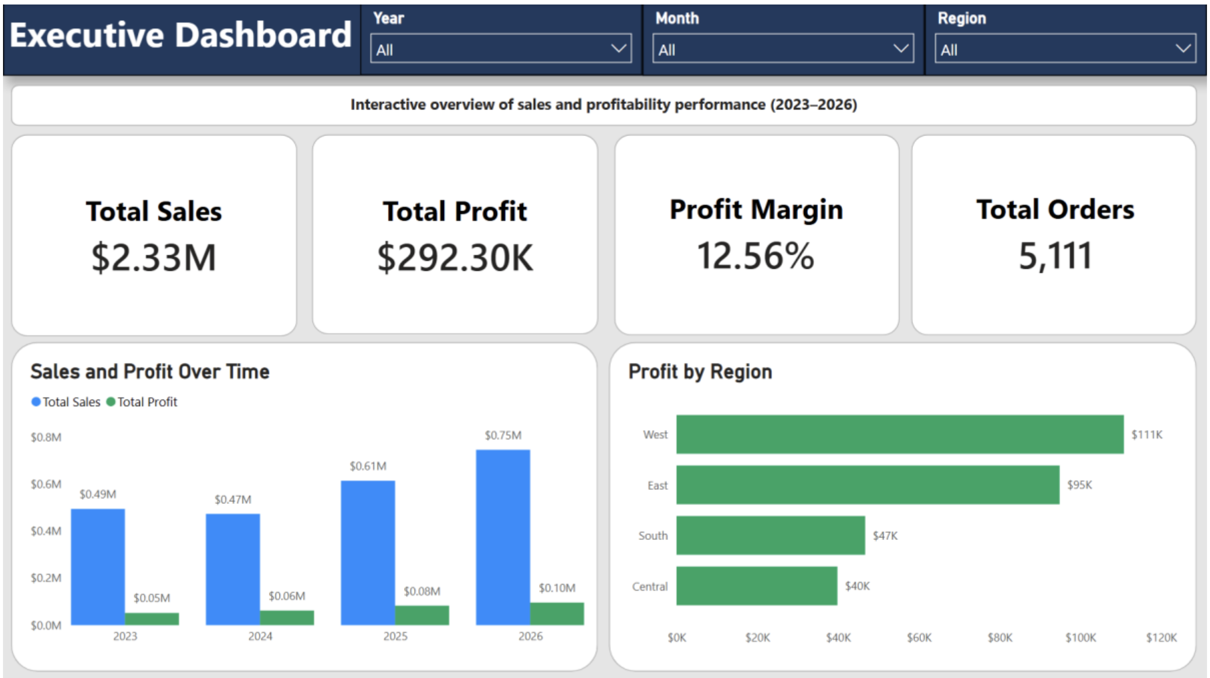

Figure B.1 Executive Dashboard

B.2 Product Performance Dashboard

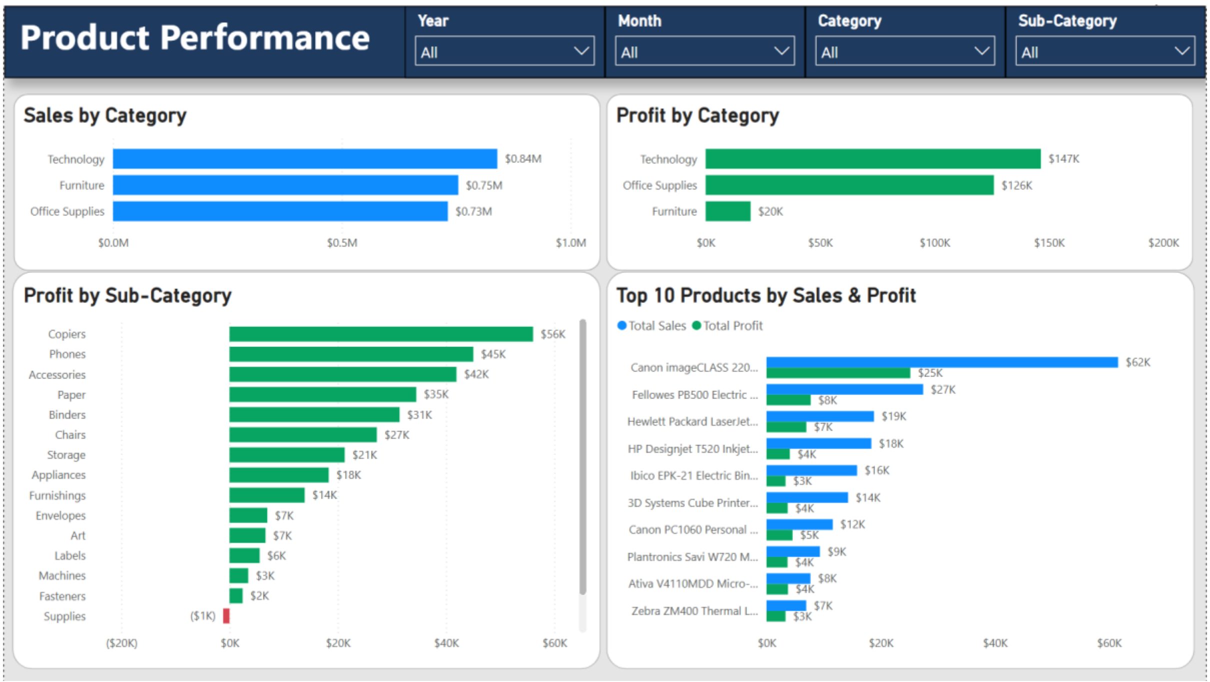

Figure B.2 Product Performance Dashboard

B.3 Customer & Regional Insights Dashboard

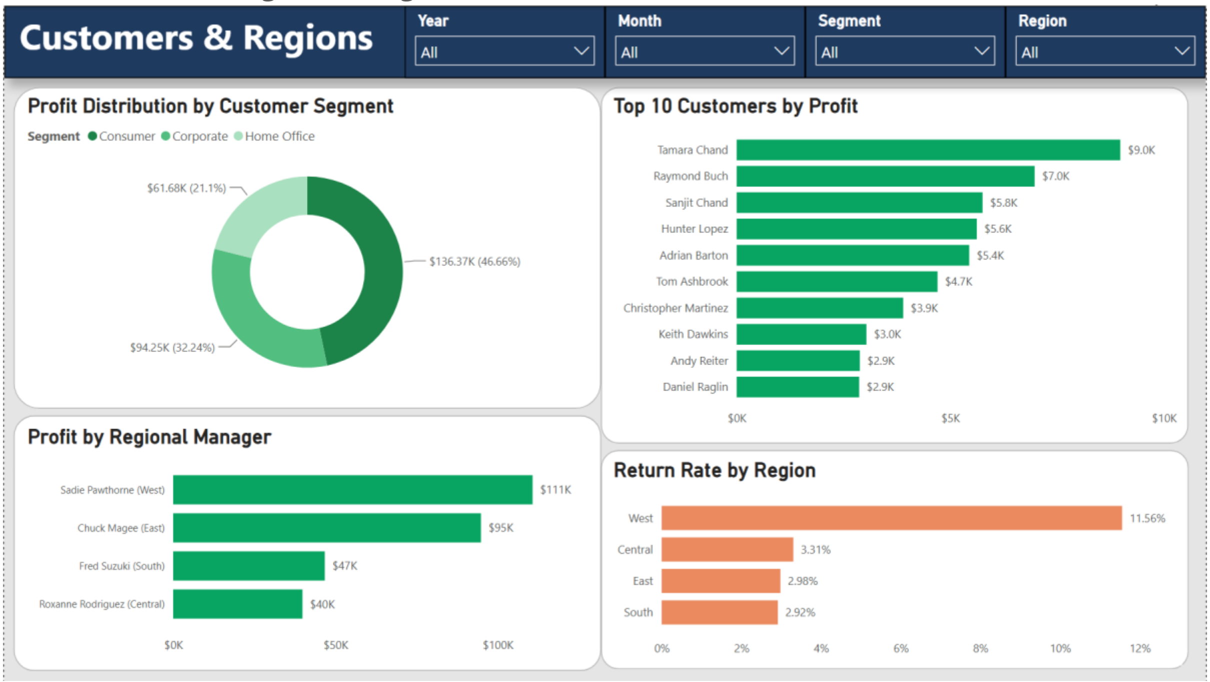

Figure B.3 Customer & Regional Insights Dashboard